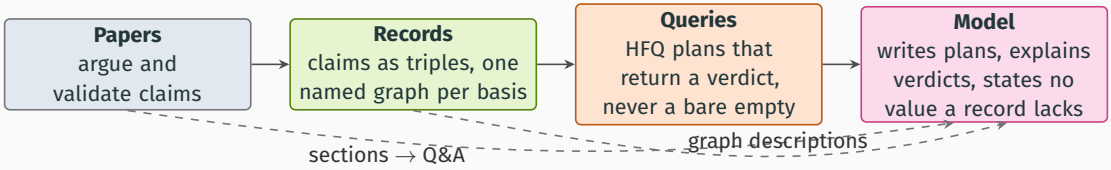
dcat-ap-plus-p450 — Components and Construction

Draft for review

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8 October 2026

1 What this is: four layers, each built from the ones beneath it

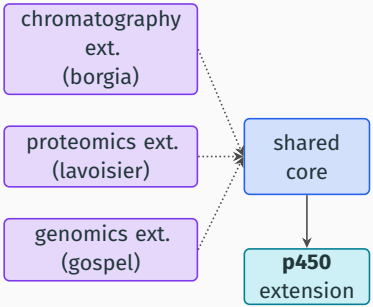


What it is not

- **Not a simulator.** Dynamics are ladders: a sequence of rungs with powers, composed as $1 - \prod_i (1 - p_i)$. The oscillator framing of the old monograph is retired.
- **Not a model that stores facts.** The model is a compiled probe: it translates questions into plans, and the plans read records.
- **Not a claim about substrate scope.** Recorded substrates are an extension; scope is an intension. The library records the first and refuses to infer the second (rule R9).

1 Why a fourth extension, and what it reuses

- The fourth DCAT-AP-PLUS extension, after **chromatography** (borgia), **proteomics** (lavoisier) and **genomics** (gospel). It reuses their machinery through a **shared core** instead of copying it a fourth time.
- The core is extracted once from chromatography (CI, typing, fifteen test modules); it starts inside this package and becomes its own package when a second extension switches to it (D2). The package layout follows proteomics, the most complete sibling.
- Schema: LinkML, importing DCAT-AP-PLUS and Chem-DCAT-AP, in its own IRI space (registration pending).
- Hosted in **Buhera OS**, which already has the CKG runtime, the vendored HFQ and ladder engines and the natural-language-to-DSL module. The library adds no runtime.



dotted: reuse of code

solid: data or control flow

Constructs used in this deck (1/2): records and their semantics

Basis	Each value is *stated* (a source or file says it), *supplied* (a convention filled it in) or *derived* (computed here). Tags combine as the maximum under stated $<$ supplied $<$ derived: a basis is never laundered (R2).
Named graph per basis	A result file keeps one named graph each for stated, supplied and derived triples, plus a *refined* graph (RDF-star refined values) and an *admitted* graph (slots admitted by schema admission).
Stage	A derived value carries the digest of the content and conventions it was computed on, and is valid only for that stage (R5).
Refined triples, gate letters	Refinement annotates asserted triples without rewriting them. The gate labels an empty answer $+$ (witnessed absent), $-$ (not recorded), a (both), `open` (a name the graph never uses) or ? (undecided).
Schema admission A1–A6	A candidate slot enters the schema only if it is demanded, witnessed from the measured object, a fresh optional predicate, from a closed vocabulary, an identity by partition, and promoted only at a federated stage.
Ground verdicts	From its PROV graph, each returned record is a *symbol* (no generation recorded), *collapsed* (generated, but no contact or derivation reachable), *index* (traced to a contact with the measured object), *icon* (traced to a derivation) or *composite* (both).

Constructs used in this deck (2/2): queries, runtime, compute

HFQ	A federated plan language (hegel). A plan is parsed and its capabilities are checked before any source is contacted; it runs within a request budget, and every step ends in a verdict.
Verdict, blocker	The label a step ends in (e.g. `starved`) and the reason for it (e.g. *upstream*), with a blame chain. Only `answered` and `empty` carry no blocker (R3).
Ladder	A sequence of rungs with powers p_i, composed as $1 - \prod_i (1 - p_i)$. Below the declared target the ladder is refused as `subfloor`, with the shortfall named.
okgg	An ontological knowledge graph generator (conspirator): a model proposes distinctions and a lexical witness decides values. Here its proposals go to schema admission only, never into records (R1).
Buhera OS	The host runtime. One registry dispatches about 40 modules, among them the CKG runtime (catalytic states as nodes; modules fold facts onto them), vendored HFQ and ladder engines, and *dsl-writer* (natural language to a DSL script, checked by the real compiler and repaired on failure).
lattice, AppHub	AppHub: the university's Code-Server sessions, CPU or GPU, accepting no inbound connection. *lattice* wraps a task into a unit carried by the university Gitea; results come back on a branch. AppHubAI is an OpenAI-compatible model endpoint there.

3 Design rules (1/3): they hold for every component

	Rule	Origin	Enforced by
R1	**The model proposes; the records decide.** No value enters a record because a model said so.	okgg, Part III	`llm.eval` (zero unbacked values); `sem.ontology` witness
R2	**Basis is never laundered.** Tags compose as max under stated $<$ supplied $<$ derived. A value computed from a supplied value is not stated.	honjo-masamune, chromatography `provenance`	`core.provenance`
R3	**No bare empty.** Every step ends in a labelled verdict; every label except `answered` and `empty` names a blocker.	HFQ, chromatography `verdicts`	`core.verdicts`
R4	**The network is touched in one place.** Only `dom.fetch` contacts external services; everything else reads dated, content-addressed snapshots.	HFQ adapters (`adapters.py:15`)	code review; `dom.fetch` is the only module importing an HTTP client

A change that breaks a rule is a design change and must be argued in a paper, not made in code.

3 Design rules (2/3)

	Rule	Origin	Enforced by
R5	**A stored value is valid only for its stage.** Every derived value carries the digest of the content and conventions it was computed on.	chromatography `identifiers.Stage`	`core.identifiers`
R6	**Every class has its own IRI.** A distinction drawn in the schema reaches the emitted RDF. Measured and computed observations are never one `rdf:type`.	chromatography schema; SGV receiver collapse	`schema.audit` (zero lost pairs)
R7	**Change upstream, then re-vendor.** Code vendored into Buhera is byte-exact and pinned; a fix is made in the upstream repository first.	Buhera spec rule M1	Buhera specs; `q.conformance`
R8	**Refined values ride alongside.** Asserted triples are never rewritten; refined values are stage-annotated annotations (RDF-star), never slots.	ART conservativity; admission tex:379	`sem.refinement`

3 Design rules (3/3)

	Rule	Origin	Enforced by
R9	**Scope is not inferred.** A substrate list is never closed under similarity to claim what an isoform *can* metabolise.	borgia `chem-dcat-ap-querying.tex:738`; ART tex:1816	`dom.chemistry`; N3 states the position
R10	**Every claim has a plan.** Each claim a paper makes ships as an HFQ plan with its expected verdict, and that pair is a test.	this specification	`q.plans`
R11	**Heavy work runs on the university servers.** Validation, catalogue builds, admission, ontology runs, training and evaluation are lattice units on AppHub.	user decision	§ 9
R12	**Every number on a front page is read from a record.** No constant is copied into a page or a module.	`/protein-modelling` thesis, made strict	`rt.p450module`, `rt.web`

4 The system at a glance (1/2): L0–L5, where things run and come from

L5 Domain modules (12)	fetch · isoforms · structures · cycle · kinetics · chemistry · metabolites · LC · abundance · PGx · spectroscopy · DDI
L4 Schema (5)	p450.yaml · refined.yaml · witness classes · OWL/SHACL/context · audit
L3 Shared core (8)	identifiers · census · refinement · verdicts · linkage · provenance · emission · artefacts
L2 Upstream code (19)	chromatography ext. · proteomics ext. · genomics ext. · HFQ · SBS · okgg · ART · honjo · CASMI · labdatareader · ladder engine
L1 Sources, snapshots only (12)	UniProt · PDB · ChEMBL · PharmVar · CPIC · ClinVar · Rhea · KEGG · FDA DDI · instrument files
L0 University compute (6)	Uni Gitea · lattice · AppHub · AppHubAI

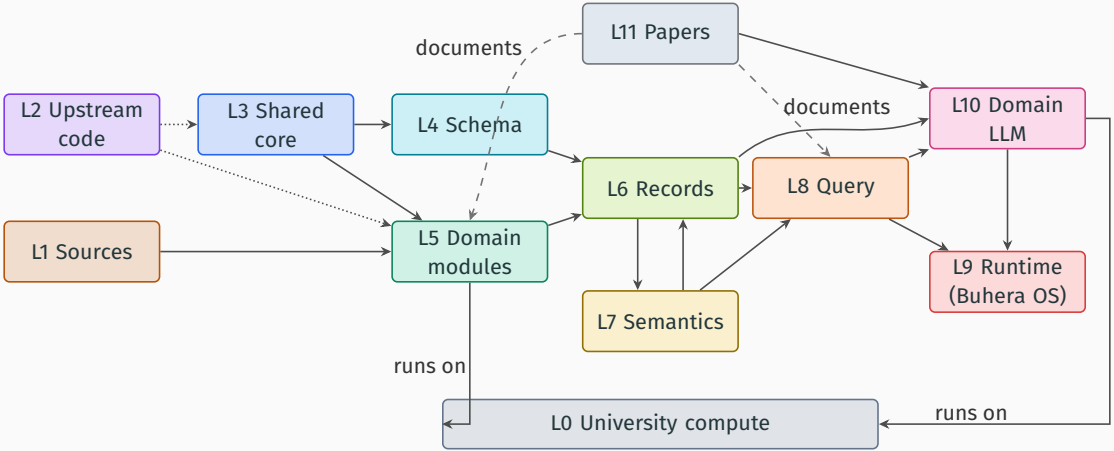
Twelve layers, stacked from where computation runs to the papers that argue for it. In brackets: components per layer in `components.json` (102 in total).

4 The system at a glance (2/2): L6–L11, what is built on them

L11 Papers (10)	ladder rewrites · records sections · N1–N8
L10 Domain LLM (5)	corpus · LoRA training · TIES merge · evaluation · serving
L9 Runtime: Buhera OS (8)	registry · CKG runtime · cytochrome module · hfq · ladder · dsl-writer · front pages
L8 Query (7)	HFQ p450 world · predicates · maps · data-derived ladders · plan library · SPARQL
L7 Semantics (4)	refined triples + gate · admission · grounding · ontology
L6 Records (6)	stated · supplied · derived · refined · admitted → result files

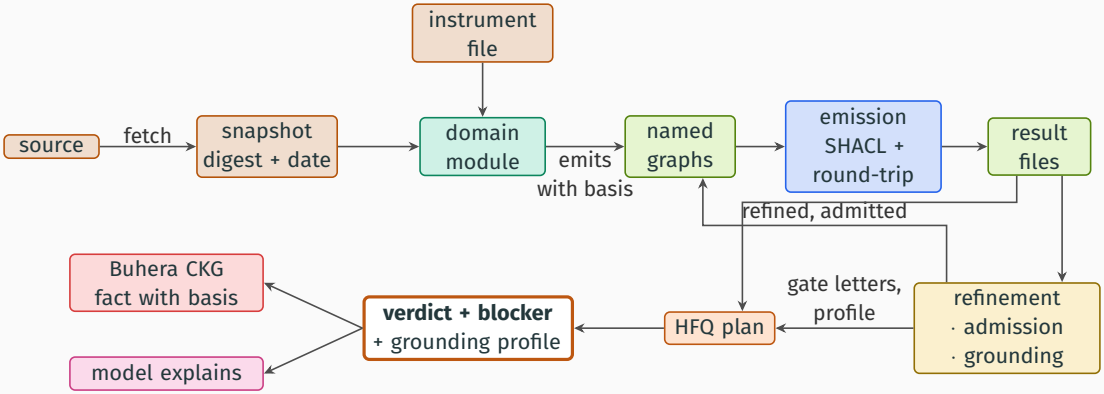
The next frame shows how the twelve layers depend on each other.

4 How the twelve layers depend on each other



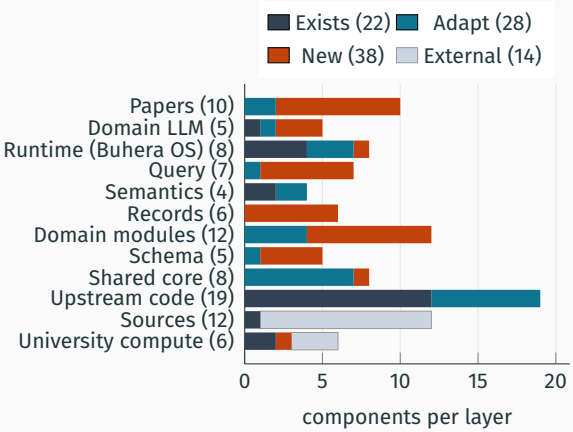
Solid: data or control flow. Dotted: reuse of code. Dashed: a paper documents the claims a component implements.

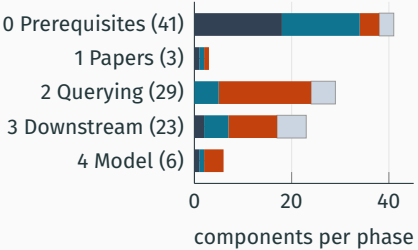
4 The single data path every value follows



Only `dom.fetch` touches the network (R4); every later step reads dated, content-addressed snapshots.

2 Status and phase of the 102 components

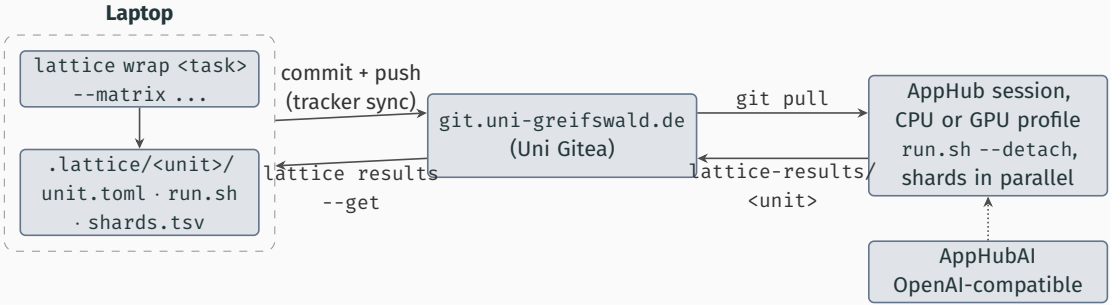




Exists: implemented and tested upstream, reused as is. **Adapt**: implemented upstream for another domain, needs P450 adaptation. **New**: exists nowhere, built here. **External**: a service or data source outside the user's repositories.

173 connections of 12 kinds: feeds 58, reuses 42, emits 17, documents 14, snapshots 10, dispatches 8, runs on 10, analyses 4, queries 4, trains 4, conforms to 1, deploys 1.

6.1 University compute: how a task reaches AppHub



AppHub accepts no inbound connection, so every task travels as a **lattice unit**: written on the laptop, pushed to the university Gitea, pulled and run on AppHub, results pushed back to `lattice-results/<unit>`.

Components: `compute.lattice`, `compute.gitea`, `compute.apphub`, `compute.apphubai`, `compute.tracker` (6: 2 exist, 3 external). § 9 lists every unit.

6.2 Sources: 12, each read only through a snapshot

Source	What is taken	First needed by	Licence note
UniProt	57 human CYPs, POR, CYB5A: sequence, features, cross-references	`dom.isoforms`	CC BY 4.0
PDB	CYP structures (e.g. 1TQN), heme site, ligands	`dom.structures`	free
Rhea	reactions for EC 1.14.14.1 and relatives	`q.world`	CC BY 4.0
KEGG	hsa00982, hsa00983, hsa00140, hsa00590	`dom.ddi`	**restricted**: do not redistribute
ChEMBL	Ki, IC50, TDI against CYP targets	`dom.kinetics`	CC BY-SA 3.0
PharmVar	star-allele definitions	`dom.pgx`	check terms
CPIC	allele function, activity scores, phenotype tables	`dom.pgx`	CC0
ClinVar	submissions incl. drug-response significance	`dom.pgx`	public domain
FDA DDI tables	index substrates, inhibitors, inducers by strength	`dom.ddi`	public domain
Spectral libraries	reference MS/MS of drugs and metabolites	`dom.metabolites`	per library
Instrument files	UV-vis, LC-MS, HPLC raw data	`dom.spectroscopy`, `dom.metabolites`	lab-owned
Monograph corpus	the 21 existing papers and their validation	papers	own work

Snapshots are dated, digest-named, never edited. DrugBank, PharmGKB left out: licences restrict redistribution.

6.3 Upstream code: 19 modules reused (12 as is, 7 adapted)

Repository	Module → reused by
borgia	chromatography ext. → core identifiers, census, refinement, linkage, provenance, verdicts; `dom.chromatography`
	honjo-masamune[a] → `dom.chemistry`
lavoisier	proteomics ext. → core emission, artefacts; `schema.refined`; `dom.abundance`; `sem.admission`
	CASMI + matchms[a] → `dom.metabolites`
gospel	genomics ext. → core verdicts, artefacts; `dom.pgx`; `sem.grounding`
others	raw2dcat[a] (raw2dcat-converter) → `dom.chromatography`; LoRA + TIES recipe (doerr-lab-model) → `llm.train`; corpus builder[a] (nfdi4cat-model) → `llm.corpus`
hegel	HFQ → core verdicts; `q.world`; `q.conformance`; `rt.hfqmodule`
	HFQ biocat[a] → `q.world`, `q.predicates`
	SBS[a] → `dom.ddi`
	probe panel → `dom.spectroscopy`, `dom.cycle`
conspirator	DCAT-AP-PLUS → `schema.p450`; ART store + gate → `sem.refinement`; okgg → `sem.ontology`
ultraviolet	Chem-DCAT-AP → `schema.p450`; labdatareader → `dom.spectroscopy`
levinthal	ladder engine → `dom.cycle`, `q.ladders`, `rt.laddermodule`; levinthal-core[a] → `dom.abundance`

[a] adapt: rewritten for P450 in this package, upstream file named in `PROVENANCE.md`. Targets: the *reuses* edges of `components.json` (10 repositories).

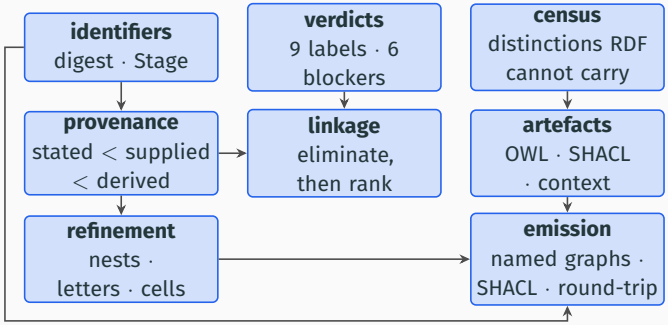
6.3 Known upstream defects this package must not inherit (1/2)

Upstream	Defect	Effect on P450	Where fixed
levinthal-core	U-lev-2: residue masses are average masses labelled monoisotopic, ∼18.01 Da heavy each	peptide masses for CYP abundance would be wrong	upstream, phase 0
levinthal-core	U-lev-3: folding unseeded, fails on wasm32; U-lev-4: `to_sentropy` constant	not used by P450 until fixed	upstream, phase 0
ladder engine / HFQ	paper says `subfloor` and `empty`; code says `starved` and raises `ParseError`	every ladder verdict would contradict its paper	upstream, phase 0 (§ 7)
genomics `corpora/clinvar.py:121`	“drug response” collapses to “O”	loses the pharmacogenomic signal	upstream, phase 0
gospel `pharmacogenetics/analysis.py:161`	`KeyError`; averages alleles instead of summing	unusable	not reused; `dom.pgx` is new

6.3 Known upstream defects this package must not inherit (2/2)

Upstream	Defect	Effect on P450	Where fixed
lavoisier `MSAnnotator`	isotope model returns empty silently; test imports a missing module	no isotope evidence for Fe or halogens	in `dom.metabolites`
lavoisier CASMI	element set lacks F, Br, I, Fe	most drugs and heme unassignable	in `dom.metabolites`
honjo-masamune	d-block out of scope	no iron chemistry	out of scope for `dom.chemistry`; heme handled by `dom.cycle` and `dom.spectroscopy`
raw2dcat-converter	`kg/`, `federation/`, `individuate/`, `analysis/` hold only `__pycache__` on `main`	that code is not reusable until its source is found	ask the user (D9)
HFQ	three copies (hegel Python, hegel JS, borgia JS)	answers could differ by runtime	`q.conformance`, phase 0

6.4 Shared core: eight modules, extracted once



- Extracted from `dcat-ap-plus-chromatography`, which has CI, typing and fifteen test modules.
- Starts as `src/dcat_ap_plus_core/` inside this package, with a `PROVENANCE.md` naming the upstream commit and every change since.
- Promoted to its own package when a second extension switches to it (D2).
- 8 modules: 7 adapted, 1 new (the verdict algebra of § 7).

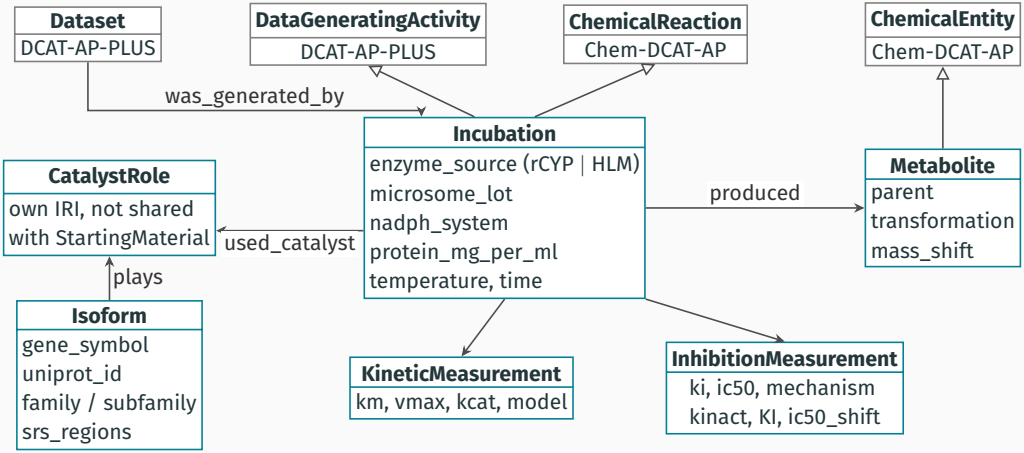
6.4 Shared core: public surface and acceptance (1/2)

Module	Public surface (sketch)	Acceptance
`identifiers`	`digest(obj) -> str`; `content_digest(bytes) -> str`; `Stage(content, conventions)`	identical inputs give identical digests across runs and machines; a changed convention changes the stage
`provenance`	`Basis` enum; `combine(*bases) -> Basis` (max)	no function returns a lower basis than its inputs
`refinement`	`refine(graph, order) -> dict[triple, Refined]`; `Nest`, `PairStatus`	conservativity: projecting refined values away returns the graph exactly
`verdicts`	`Label`, `Blocker`, `Evidence`, `assign(evidence) -> Verdict`; `from_hfq(step) -> Verdict`	every HFQ verdict maps to exactly one label (§ 7); a label without a blocker is only `answered` or `empty`

6.4 Shared core: public surface and acceptance (2/2)

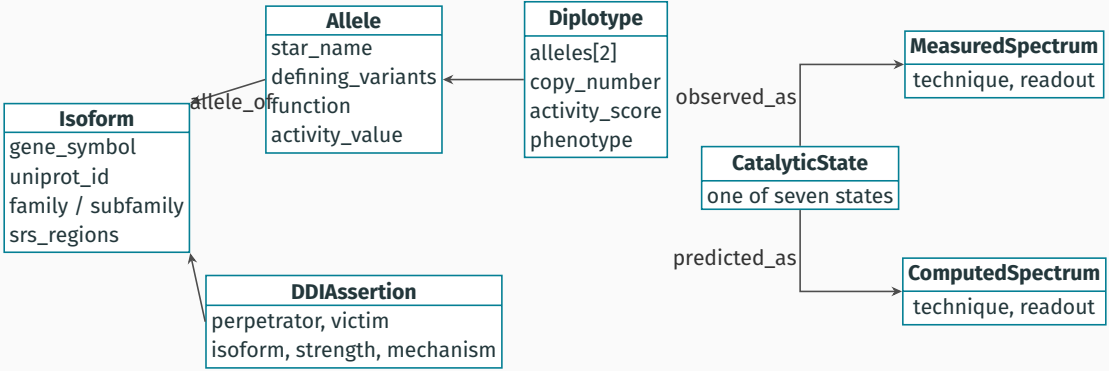
Module	Public surface (sketch)	Acceptance
`linkage`	`link(records, bases) -> list[Link]`	no returned link is contradicted by a stated field
`census`	`census(schema_view) -> Census`	reports class/slot fibres sharing an IRI and the pairs they lose
`artefacts`	`generate(schema) -> {owl, shacl, context}`	generated SHACL validates the synthetic fixtures; includes the genomics SHACL repairs
`emission`	`emit(records, schema) -> Dataset` (rdflib); `round_trip(file) -> bool`	every result file conforms and parses back to the same quads

6.5 Schema: class sketch (1/2), incubations



A proposal for N1. The final classes are decided by the census and the receiver audit, not by this diagram.

6.5 Schema: class sketch (2/2), genotypes, states, interactions



Measured and computed spectra are separate classes on purpose (next frame).

6.5 Schema: two decisions fixed by the design rules

- **`CatalystRole` has its own IRI** (R6).
  In Chem-DCAT-AP, `Catalyst` and `StartingMaterial` share the exact mapping `VOC4CAT:0000194`. Under that mapping a query cannot tell an enzyme from a substrate.
- **`MeasuredSpectrum` and `ComputedSpectrum` are distinct classes** (R6).
  The semiotic-verdicts paper shows that six activity classes emit as `prov:Activity` and become indistinguishable. The remedy is a fibre-unique `rdf:type`.

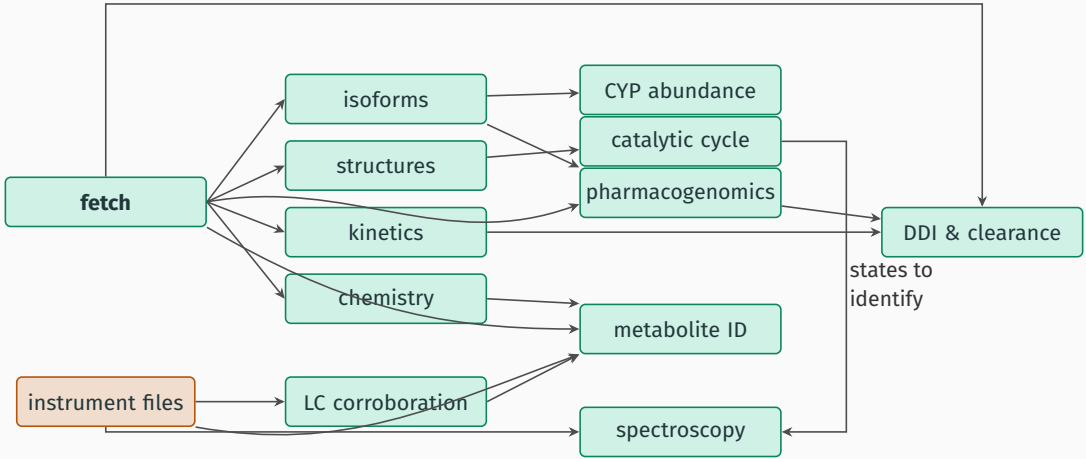
Both are instances of R6: a distinction drawn in the schema must reach the emitted RDF. `schema.audit` checks it (zero lost pairs), and WP2.1 is done only when measured and computed spectra keep distinct IRIs.

6.5 Schema: witness classes, so that grounding can be computed

`schema.witness` declares which activities count as contact with the measured object and which as derivation. The grounding queries of § 7.3 read these sets.

Witness set	Proposed members
A_c contact activities	`Incubation`, `SpectralAcquisition`, `LCMSAcquisition`, `Genotyping`
U_c contact inputs	`Instrument`, `MicrosomeLot`, `RecombinantEnzyme`
A_d derivation activities	`MetaboliteIdentification`, `PhenotypeAssignment`, `LadderEvaluation`, `SpectrumComputation`
U_d derivation inputs	`Snapshot`, `ReferenceSpectrum`
G_d derivation agents	`prov:SoftwareAgent`

6.6 Domain modules: how data moves between them



Each module reads snapshots and instrument files and emits records with a basis; none of them answers questions.

6.6 Domain modules (1/2)

Module	Input → output	Basis	Acceptance
`dom.fetch`	source query → snapshot dir + manifest	—	re-fetch with no upstream change gives the same digest; no other module imports an HTTP client
`dom.isoforms`	UniProt snapshot → 57 isoforms, families, SRS, CYP2D6/2D7 pair	supplied	count = 57; SRS positions per Gotoh 1992 on CYP2 family
`dom.structures`	PDB/GLB → heme site, access channels	supplied	Fe position of 1TQN reproduces the GLB pipeline value
`dom.cycle`	seven states, rung powers → ladder, composite, `subfloor`	derived	ladder tests incl. both refusals of `def:refuse-climb`
`dom.kinetics`	ChEMBL + incubations → Km, Vmax, Ki, IC50, kinact/KI with units and conditions	supplied / stated	two values compared only under declared tolerances; undecided where either side is supplied
`dom.chemistry`	SMILES → structure with stated/supplied tags, candidate sites	supplied / derived	never asserts scope (R9)

6.6 Domain modules (2/2); 12 in all, 8 new, 4 adapted

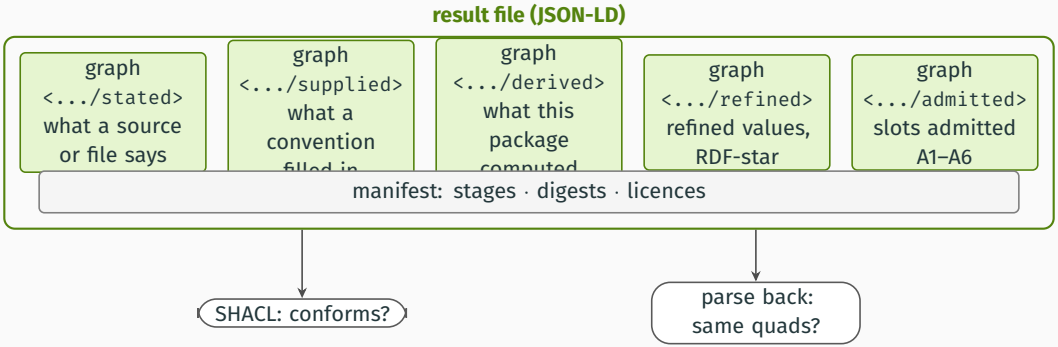
Module	Input → output	Basis	Acceptance
`dom.metabolites`	spectra + parent → metabolite ID or decline	stated + derived	licensed only when mass rule and spectral match agree; every decline has a reason
`dom.chromatography`	LC runs → parent/metabolite peak fields	stated	retention corroborates, never identifies
`dom.abundance`	surrogate peptides → isoform abundance	derived	peptides mapped by sequence, not by label
`dom.pgx`	genotypes + PharmVar/CPIC → diplotype, activity score, phenotype	supplied / derived	activity score summed, never averaged; CPIC tables reproduced exactly
`dom.spectroscopy`	UV-vis now, EPR/rR/Mössbauer new → state assignment	stated + derived	minimal probe set of `exp_04_panel` reproduced
`dom.ddi`	kinetics + PGx + pathways → DDI assertions	derived	FDA index pairs recovered with stated strength

6.6 Biotransformation rules used by metabolite identification

Transformation	Mass shift (Da)	Notes
hydroxylation / N-, S-oxidation	+15.9949	the most common P450 product
demethylation (N-, O-)	−14.0157	dealkylation family
dehydrogenation / desaturation	−2.0157	atypical reactions atlas
epoxidation	+15.9949	isobaric with hydroxylation; needs MS/MS
hydroxylation + dehydrogenation	+13.9793	ketone formation
glucuronidation (downstream)	+176.0321	phase II, flagged as not P450
GSH adduct (reactive metabolite)	+305.0682	with neutral loss −129.0426

How the rules are applied, and when the module declines: flow F3 (§ 8).

6.7 Records: named graphs, one per basis, in one result file per dataset



The layout follows proteomics, which already emits 34 files and 4.9 million quads this way. Asserted triples are never rewritten: refinement and admission add their own graphs (R8).

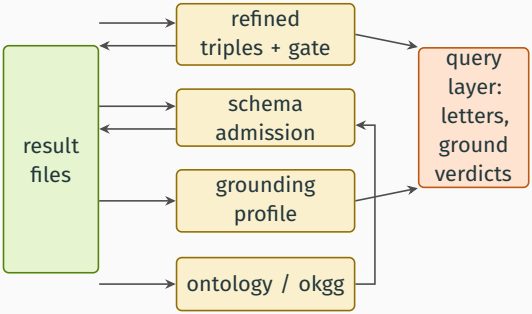
6 record components (stated, supplied, derived, refined and admitted graphs; result files), all new.

6.7 A record in the derived graph (TriG sketch)

```
GRAPH <https://w3id.org/nfdi-de/dcat-ap-plus/p450/run/2026-10/derived> {
  p450:met/7f3a  a p450:Metabolite ;
      p450:parent        chebi:CHEBI_41879 ;          # midazolam
      p450:transformation p450:Hydroxylation ;
      p450:massShift     "15.9949"^^xsd:decimal ;
      prov:wasGeneratedBy p450:act/MetaboliteIdentification/91c2 .
  p450:act/MetaboliteIdentification/91c2
      a p450:MetaboliteIdentification ;
      prov:used p450:snap/massbank/2026-10-08/3be1... ;
      prov:wasAssociatedWith p450:agent/dcat-ap-plus-p450/0.1.0 .
}
```

The PROV chain names a `MetaboliteIdentification` activity, a snapshot and a software agent: members of A_d, U_d and G_d in the witness classes, which is what the grounding queries read (§ 7.3).

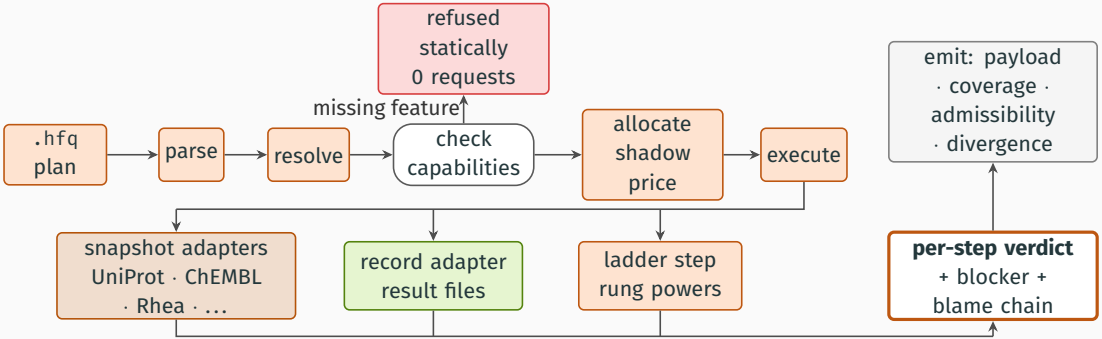
6.8 Semantics: four analyses that add to records without changing them



Paired arrows: the refined and admitted graphs are written back. The arrow on the right carries okgg's proposed slots to admission.

- **Refined triples and the gate** (`sem.refinement`). An empty answer is labelled + (witnessed no), − (not recorded), a (both) or `open` (a name the graph never uses).
- **Schema admission** (`sem.admission`). Fields are scored by the pairs they tell apart; cells the schema cannot split are where it learns. Slots are admitted only under A1–A6.
- **Grounding profile** (`sem.grounding`). The five ground verdicts per record, reported beside every answer.
- **Ontology** (`sem.ontology`). okgg facets over papers, sections and records; its proposals go to admission, never into the records directly (R1).

6.9 Query: HFQ with a p450 world



New predicates (`q.predicates`, added to `PREDICATE_FEATURES` following the `biocat.py:95` pattern): `inhibits(?c, ?isoform)`, `induces(?c, ?isoform)`, `substrate_of(?c, ?isoform)`, `metabolite_of(?m, ?parent)`, `allele_of(?a, ?isoform)`, `activity_score(?diplotype)`. HFQ `filter` today allows one comparison and no units; numeric range filtering with units is a language extension. The world is added upstream in hegel (R7), validated there, and re-vendored into Buhera.

6.9 An illustrative plan and the verdicts it is expected to return

```
plan cyp3a4_potent_inhibitors_on_fda_list {
  budget 300 requests
  let inh   = from CHEMBL
      ask inhibits(?c, "CYP3A4")
      within 120
  let potent = filter inh where ki_uM < 1.0
  let ids   = map potent via chembl_to_chebi
      expect partial 0.8
  let fda   = from FDA
      ask strong_inhibitor_of(?c, "CYP3A4")
      with ?c in ids
      within 60
      when starved emit partial
  emit fda with provenance
}
```

In the current grammar. The `filter` on a unit-bearing attribute and the `FDA` source need `q.predicates` and `q.world`.

Expected verdicts (the test `q.plans` keeps beside it):

- `inh`: answer
- `potent`: answer
- `ids`: answer with retention ≥ 0.8, or `starved` naming the map if retention falls below 0.8
- `fda`: answer or partial

An empty `fda` is reported with a gate letter, not as “none”.

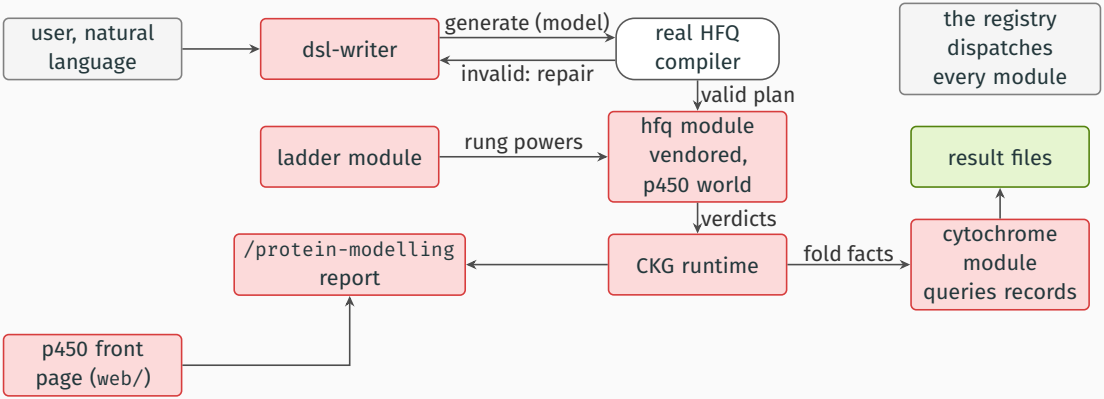
6.9 A ladder over the seven-state cycle, and why it is refused

```
plan cyp3a4_cycle_coupling {
  budget 20 requests
  let states = from P450REC
      ask catalytic_states("CYP3A4")
      within 10
  let cycle  = ladder over states power 0.62, power 0.55, power 0.48, power 0.41
               expect power 0.95                    (one line in the plan; wrapped here)
  emit cycle with provenance
}
```

$$1 - \prod_i (1 - p_i) \;=\; 1 - (0.38 \cdot 0.45 \cdot 0.52 \cdot 0.59) \;=\; 0.9475 \;<\; 0.95$$

- By `def:refuse-climb` the ladder is refused with `subfloor`, shortfall 0.0025. Today's code returns `starved` here; § 7.1 fixes that.
- The data-derived extension (`q.ladders`) replaces the literals with powers read from records, or from the ladder engine's computed intensive power.

6.10 Runtime: Buhera OS. The library adds no runtime



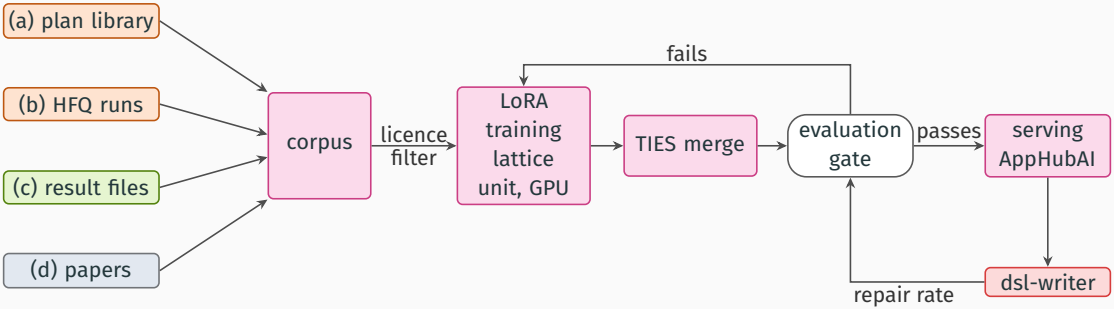
Buhera already has the registry (about 40 modules), the CKG runtime, the vendored HFQ and ladder engines and dsl-writer. The library supplies records, a world, plans and a model to it.

8 runtime components: 4 exist, 3 adapted, 1 new (the front page).

6.10 Runtime: what changes in Buhera, in order

1. Re-vendor HFQ with the `p450` world and the ladder fixes (R7).
2. Rewrite `cytochrome-module.js` so that each op queries the records instead of holding constants (R12). Decision D7: rewrite in place, or keep the current module as the transaminase-comparison demo and add a new `p450` module.
3. Give every CKG fact its basis and ground verdict. The current report mixes measured literature values (Soret 417$\rightarrow$392 nm, the EPR g-tensor, Raman 795$\rightarrow$758 cm^{-1}), framework-derived values (ΔM, $\Sigma\Delta M = 4.963$, the conditioned floor β), and framework values that read as clinical (CYP2D6 phenotype ΔM is not a CPIC activity score). After this change the report can say which is which.
4. Point dsl-writer at the domain model for `dslId: "hfq"`.

6.11 Domain LLM: a compiled probe, trained only on what the library produced



Corpus tasks: (a) question → plan; (b) plan → verdict → explanation, refusals included; (c) graph → description; (d) section → Q&A.

The model translates questions into HFQ plans and explains verdicts; it is never a source of values (R1). It deploys behind a compiler that rejects any plan it gets wrong: dsl-writer's check-and-repair loop already exists.

6.11 Domain LLM: what the corpus contains

Corpus task	One JSONL line	Example output, abridged from the specification
`plan`	question → plan, with its source `.hfq` file	`plan cyp3a4_potent_inhibitors_on_fda_list { ... }`
`explain`	plan + per-step verdicts → explanation	"The ChEMBL-to-ChEBI map kept 71% of the inhibitors, below the declared 80%, so the FDA step was not issued. The empty result says nothing about the FDA list."
`refuse`	question → refusal	"The library records substrates observed in incubations; it does not infer scope (rule R9)."

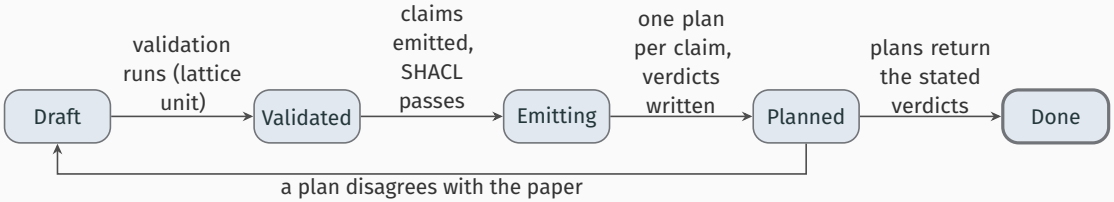
Corpus records are JSONL, one task per line; refusals are included. The corpus is licence-filtered before training (WP4.1).

6.11 Domain LLM: evaluation measures five things, none of them text similarity

Metric	Definition	Gate
parse-and-check rate	generated plans that parse and pass the capability check	≥ 0.95
verdict agreement	generated plan gives the same per-step verdicts as the reference plan	≥ 0.90
refusal accuracy	refuses exactly when the reference refuses	≥ 0.95
unbacked values	values stated that no record holds	**0**
repair rate	fraction of dsl-writer calls needing a repair	tracked in deployment

Baseline Qwen2.5 0.5B with LoRA (lattice unit `train`, seed $\times$ size matrix); size decided after evaluation (D6). 5 model components: 3 new, 1 adapted, 1 existing.

6.12 Papers: when a paper is finished



A paper is finished when four things hold:

1. its validation runs;
2. its records emit and conform;
3. its plans return the verdicts it states;
4. its “Records and queries” section lists what the schema cannot yet separate.

Every transition except the first is checked by a lattice unit (F10).

6.12 Papers: the work, by group

Group	Papers	Work
Ladder rewrites	expression algebra (40 oscillator refs), multi-hop ET (22), address manifold (22), resting/bound states (18), Compound I (6), introduction (5)	replace oscillator dynamics by ladders; expression algebra first, because the others cite it
Records sections	the other 15	add “Records and queries”
Merges	the two 57-isoform papers; database recovery + empty dictionary	one paper each
New	N1 schema · N2 cycle as ladder · N3 federated querying · N4 metabolite ID · N5 PGx admission · N6 DDI propagation · N7 spectroscopic discrimination · N8 domain model	
Last	the introduction	written when everything it introduces exists

10 paper components: 8 new (N1–N8), 2 adapted (ladder rewrites, records sections).

7.1 Step verdicts: one canonical algebra (proposal)

Canonical label	Blocker	Rule (first that applies)	HFQ today
`unsupported`	capability	a required feature is not declared	`surface` (model)
`starved`	upstream	an input step did not answer	`starved` (corpus)
`refused`	budget	the budget does not cover the request	`refused` (budget)
`exhausted`	budget or engine	halted during the request	`timeout` (engine)
`malformed`	record	the record does not parse	—
`subfloor`	floor	composite power below the declared target	`starved` (wrong)
`empty`	—	completed; nothing matches	`empty`
`underdetermined`	convention	an element has several readings	—
`incomplete`	record	a requested feature is absent	—
`answered`	—	none of the above	`answer`

- Proposal: the nine ordered labels and six blockers of chromatography's `verdicts.py` become canonical (a strict superset of HFQ's six), plus `subfloor` for ladders, as the ladder paper says.
- HFQ's `starved` also fires when a lossy map's retention falls below its declared expectation; canonically that is `incomplete` on the map and `starved` on the *next* step.
- The ladder paper refuses a rungless ladder with `empty`, but that ladder never ran. D8 keeps `empty` by default; the alternative is `malformed`.

7.2–7.3 Two axes orthogonal to the step verdict

Empty-answer letters

§ 7.2, from the gate of `sem.refinement`

An `empty` step is further labelled:

+	witnessed absent
−	not recorded
a	both
open	a name the graph never uses
?	undecided

Ground verdicts

§ 7.3, per returned record, from its PROV graph

Verdict	(g, c, d)	Meaning
symbol	000	no generation recorded
collapsed	100	generated, but neither contact nor derivation reachable
index	110	traced to a contact with the measured object
icon	101	traced to a derivation
composite	111	both

g: a generation recorded; c, d: a contact or a derivation reachable along `CHAIN = prov:wasGeneratedBy/(prov:used/prov:wasGeneratedBy)*`.

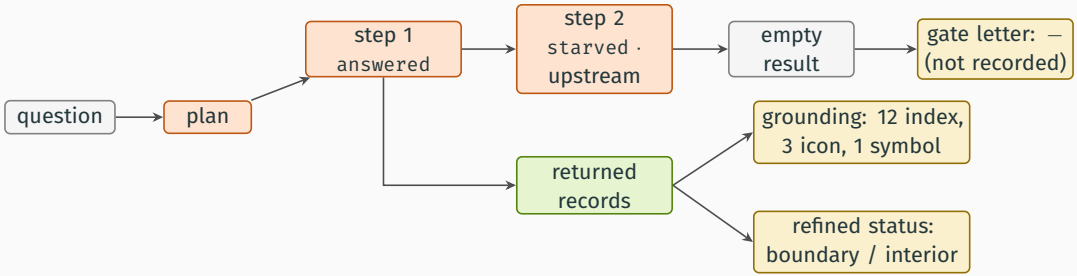
Four kinds of verdict exist upstream. They answer different questions and are kept as separate axes.

7.3 The three grounding queries

```
SELECT DISTINCT ?r WHERE { ?r prov:wasGeneratedBy ?a }                    # Q_g
SELECT DISTINCT ?r WHERE { ?r CHAIN ?a .
  { ?a a ?t FILTER(?t IN (A_c)) } UNION
  { ?a prov:used ?u . ?u a ?t FILTER(?t IN (U_c)) } }                      # Q_c
SELECT DISTINCT ?r WHERE { ?r CHAIN ?a .
  { ?a a ?t FILTER(?t IN (A_d)) } UNION
  { ?a prov:used ?u . ?u a ?t FILTER(?t IN (U_d)) } UNION
  { ?a prov:wasAssociatedWith ?g . ?g a ?t FILTER(?t IN (G_d)) } }         # Q_d
```

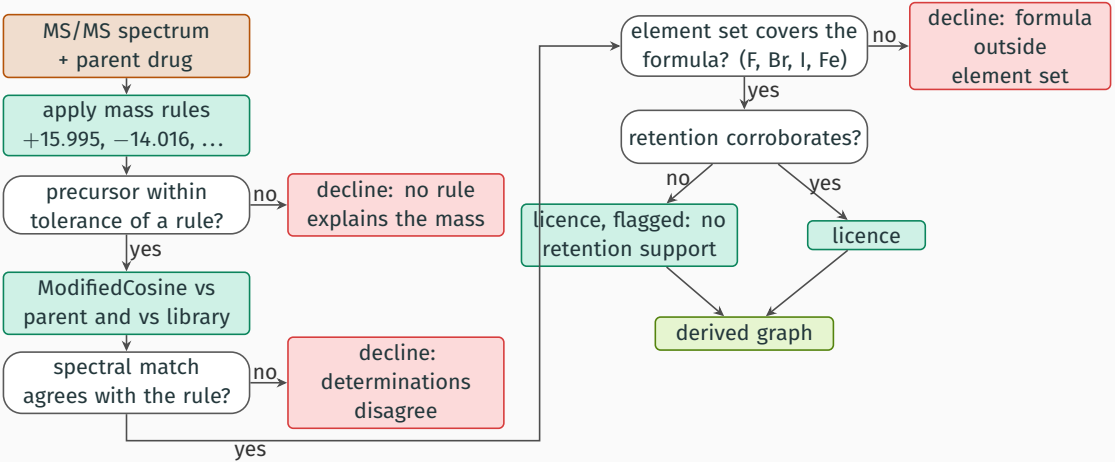
A_c, U_c, A_d, U_d, G_d are the witness classes of § 6.5. Membership in Q_g, Q_c, Q_d gives the bits (g, c, d) of the previous frame. WP3.11 runs these queries through the SPARQL endpoint.

7.4 One answer, all axes



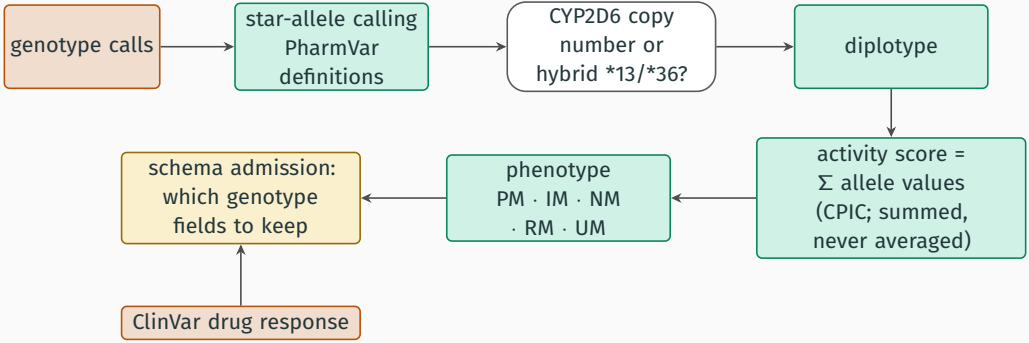
An answer the library returns therefore carries: the step verdicts with blockers and the blame chain; for empty steps, a gate letter; for each record, a ground verdict and a refined status; and the coverage and admissibility HFQ already emits.

F3 Metabolite identification: licence only when the evidence agrees



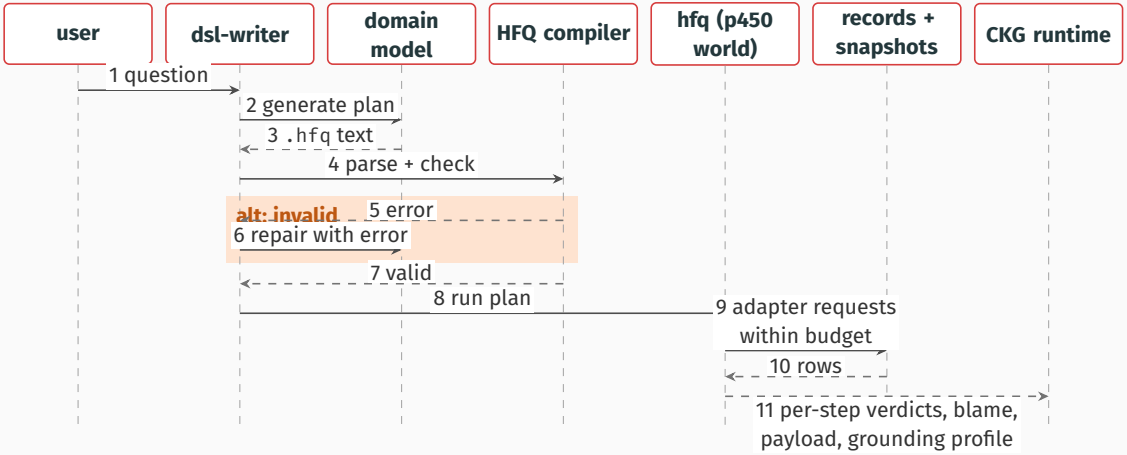
Every decline is written as a **decline record with its reason**; licensed identifications go to the derived graph. Retention corroborates, never identifies. The element set and the isotope model repair the lavoisier defects (§ 6.3).

F4 Genotype to phenotype



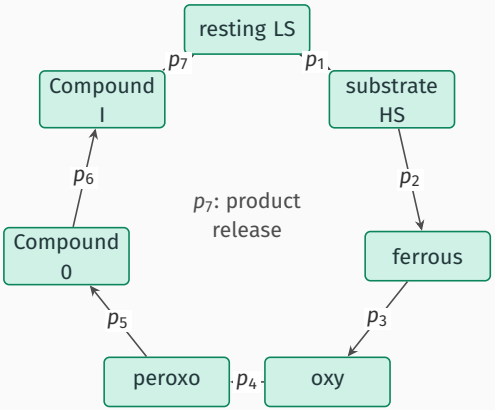
The activity score is the sum of allele values; gospel's existing analysis averages them and is not reused (§ 6.3). ClinVar's drug-response significance is kept, which needs the upstream fix to `clinvar.py:121` (WP0.7). Done when the CPIC tables are reproduced exactly (WP3.7, N5).

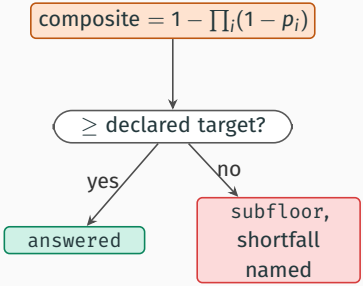
F5 A question answered through Buhera



(1) “Which potent CYP3A4 inhibitors are on the FDA list?” (12) facts returned to the user, folded with their basis, explained by the model.

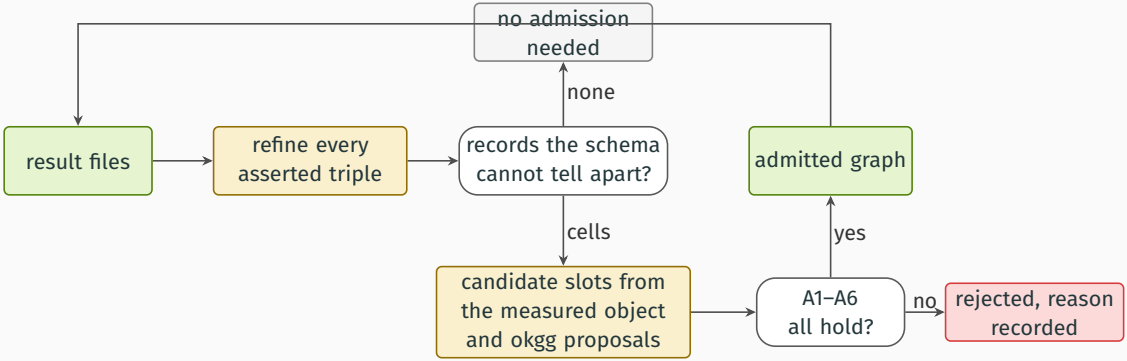
F6 The catalytic cycle as a ladder





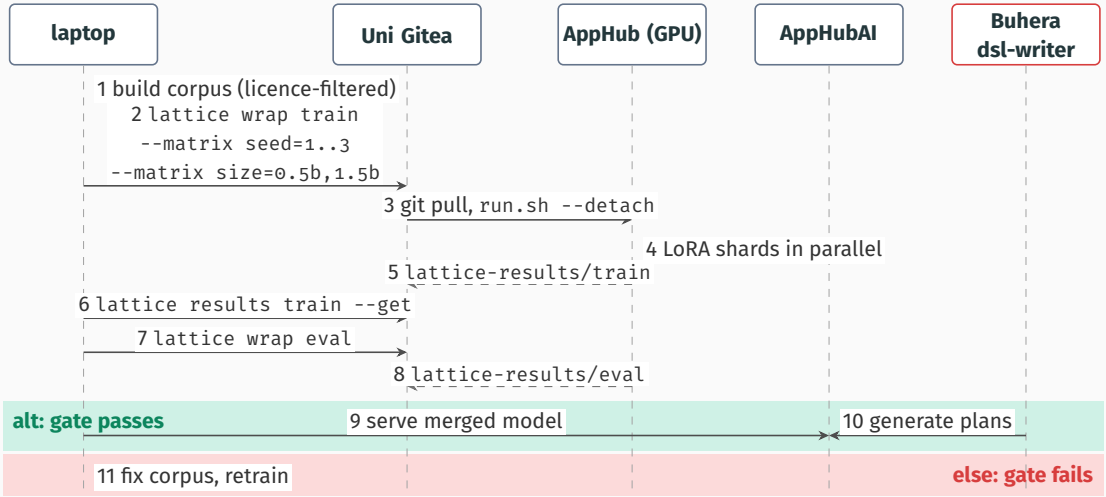
Dynamics are ladders, not oscillators. N2 argues it (WP1.1); `dom.cycle` computes it; the HFQ ladder step, later with data-derived powers (`q.ladders`), queries it.

F7 Refinement and admission



A1 demanded · A2 witnessed from the measured object · A3 a fresh optional predicate · A4 a closed vocabulary · A5 identity by partition · A6 promotion only at a federated stage. Admitted slots are reported beside random controls (WP3.2).

F9 Training and deploying the model



9.1 Computing on the university servers: every heavy task is a lattice unit

Unit	Command	Matrix	Profile	Outputs
`fetch`	`dcat-ap-plus-p450 fetch {source}`	`source=uniprot,pdb,rhea, chembl,cpic,clinvar,fda`	CPU	`snapshots/**`
`validate`	each paper's `validation/run_all.py`	`paper=...` (21)	CPU	`validation/ results/**`
`catalogue`	`dcat-ap-plus-p450 catalogue --validate`	`--each 'results/inputs/*.json'`	CPU	`results/ catalogue/**`
`admit`	`dcat-ap-plus-p450 admit`	—	CPU	`results/ admission/**`
`ontology`	`okgg run ... --unit section`	`paper=...`	CPU + AppHubAI	`results/ ontology/**`
`conformance`	Python ↔ JS HFQ on every plan	—	CPU	`results/ conformance.json`
`train`	`python model/train/train.py`	`seed=1..3, size=0.5b,1.5b`	GPU	adapters
`eval`	`python model/eval/eval.py`	`model=...`	GPU	`results/ model-eval/**`

`lattice plan <task>` prints what a unit needs before anything is written. The route: § 6.1.

9.3 What may be pushed where

Path	Public GitHub	University Gitea	Reason
`src/`, `tests/`, `docs/`, `web/`, `plans/`, `artefacts/`	yes	yes	own work
`snapshots/`: uniprot, pdb, rhea, chembl, cpic, clinvar, fda	yes	yes	open licences (ChEMBL is share-alike)
`snapshots/kegg`	**no**	yes, if the KEGG licence allows internal use	KEGG forbids redistribution
`snapshots/instruments`	ask the lab	yes	lab-owned
`model/corpus`	only the open-licence slices	yes	the corpus inherits its sources' licences
`.claude/`, `.env*`	**no**	**no**	secrets

The projections are kept by a tracker sync manifest (`.sync.toml`); tracker pushes each origin only what it may see.

9.2, 9.4 What the route needs first, and one known gap

Prerequisites for the route (§ 9.2)

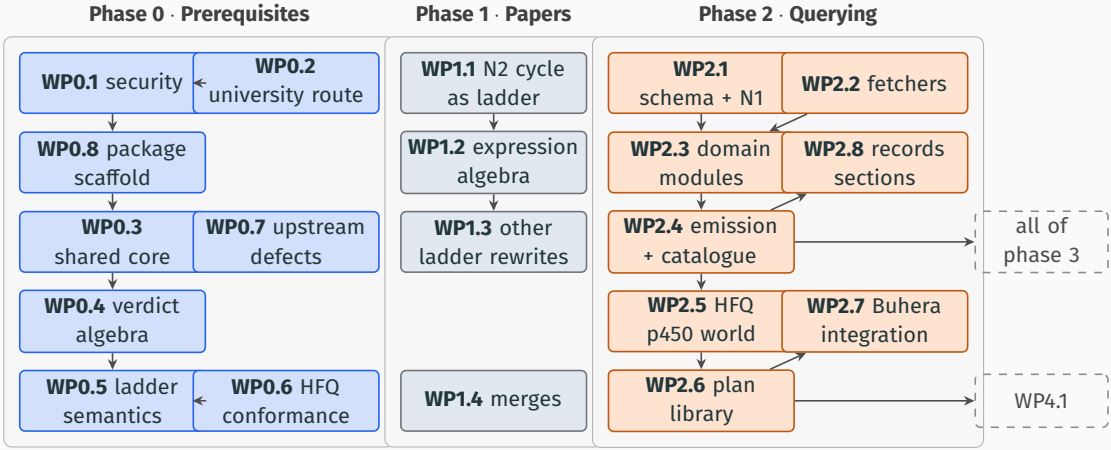
1. **A university remote.** `levinthal`'s only remote is the public GitHub origin; lattice needs a repository on `git.uni-greifswald.de`.
2. **A tracker sync manifest** (`.sync.toml`) so the university origin and the public origin each get the projection they may see. At minimum, hide `.claude/` and every `.env*` from both.
3. **A credential in pushed public history** must be rotated and its file untracked before any of this (WP0.1).

One known gap (§ 9.4)

okgg's model generator speaks to Ollama. AppHubAI is OpenAI-compatible and does not offer Ollama, and lattice flags Ollama use.

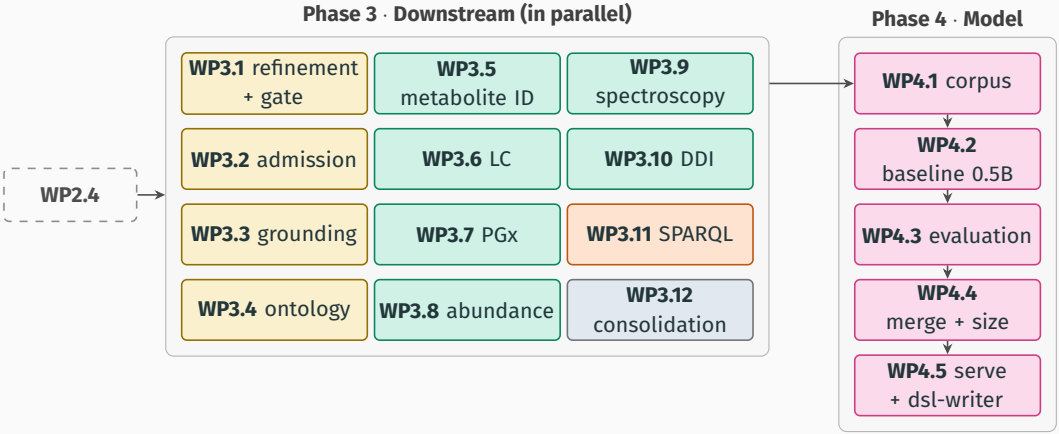
Either okgg gains an OpenAI-compatible generator (upstream in conspirator, R7), or ontology runs stay local with the lexical generator or a local Ollama.

10.1 Work packages and dependencies (1/2): phases 0–2



Across phases: WP1.1 after WP0.5; WP2.1 after WP0.3; WP2.2 after WP0.2; WP2.5 after WP0.6; WP3.8 after WP0.7 (U-lev-2).

10.1 Work packages and dependencies (2/2): phases 2–4



WP2.4 gates all of phase 3; WP3.8 also waits for WP0.7 (U-lev-2). The corpus (WP4.1) waits for the plan library (WP2.6) and for phase 3.

10.2 Phase 0 — Prerequisites (1/2)

WP	Work	Done when	Components
WP0.1	Untrack `.claude/settings.local.json`, add it to `.gitignore`, rotate the gateway credential	the file is absent from `git ls-files`; the old credential is revoked	—
WP0.2	University remote on `git.uni-greifswald.de`; `.sync.toml` hiding `.claude/`, `.env*`, `snapshots/kegg` from the public origin; `lattice plan validate` dry-run	`tracker sync status` shows the intended projections	`compute.*`
WP0.3	Extract the shared core from chromatography at a pinned commit; `PROVENANCE.md`; port its tests	all ported tests pass; mypy and ruff clean	`core.*`
WP0.4	Canonical step algebra (§ 7.1) in `core.verdicts`, with `from_hfq()`	every HFQ verdict in hegel's V1–V16 maps to exactly one label	`core.verdicts`

10.2 Phase 0 — Prerequisites (2/2)

WP	Work	Done when	Components
WP0.5	Ladder semantics: `subfloor` and the rungless case, in levinthal's ladder engine and hegel's HFQ (Python and JS), with tests; re-vendor to Buhera	both refusals of `def:refuse-climb` and `thm:nonvacuous` are tests that pass	`up.ladderengine`, `up.hfq`, `rt.laddermodule`, `rt.hfqmodule`
WP0.6	Conformance harness: every plan in hegel's `plans/` runs in Python and JS with identical verdicts; decide borgia's copy	harness green; borgia copy re-vendored or retired	`q.conformance`
WP0.7	Upstream defects: U-lev-2, U-lev-3, U-lev-4; genomics `clinvar.py:121`	each has a regression test upstream	`up.levcore`, `up.genomics`
WP0.8	Package scaffold: `pyproject.toml`, `Makefile`, CLI skeleton, CI, `docs/specification/` numbering, synthetic fixtures	`uv run pytest` passes on an empty package	—

10.2 Phase 1 — Papers

WP	Work	Done when
WP1.1	N2, the cycle as a ladder; absorbs the closed-orbit paper	validation runs; ladder tests from WP0.5 cited
WP1.2	Expression algebra: oscillator dynamics $\rightarrow$ ladders	no oscillator dynamics remain; R_{bio} unchanged; validation runs
WP1.3	Multi-hop ET, address manifold, resting/bound states, Compound I	same criteria
WP1.4	Merge the two isoform papers; merge database recovery with the empty dictionary	one paper each; the empty-dictionary paper becomes the bridge to N3

WP1.1 follows WP0.5, so that N2 cites ladder tests that pass. Expression algebra precedes the other rewrites because they cite it.

10.2 Phase 2 — Querying

WP	Work	Done when
WP2.1	`p450.yaml`, `refined.yaml`, `witness.yaml`; artefacts; census and receiver audit; N1 alongside	zero lost pairs; measured and computed spectra keep distinct IRIs
WP2.2	Fetchers for UniProt, PDB, Rhea, KEGG, ChEMBL as lattice unit `fetch`	snapshots with manifests; re-fetch without upstream change is a no-op
WP2.3	`isoforms`, `structures`, `cycle`, `kinetics`, `chemistry`	each module's acceptance in § 6.6
WP2.4	Emission and catalogue (lattice unit `catalogue`)	every file conforms and round-trips
WP2.5	HFQ `p450` world, predicates, maps, data-derived ladders, upstream in hegel	hegel validation extended with P450 checks, all hold
WP2.6	Plan library: one plan per claim, expected verdicts as tests	every plan returns its expected verdict
WP2.7	Buhera: re-vendor, `cytochrome-module` reads records, CKG facts carry basis	`/protein-modelling` renders with no constant copied from a paper
WP2.8	"Records and queries" sections in the 15 papers	each section lists its triples, plans and cells

10.2 Phase 3 — Downstream (1/2): semantics

WP	Work	Done when
WP3.1	Refinement and gate over the catalogue	conservativity holds; empty answers carry letters
WP3.2	Admission for P450 (lattice unit `admit`)	admitted slots satisfy A1–A6, beside random controls
WP3.3	Grounding profile beside every answer	Q_g, Q_c, Q_d run on the catalogue
WP3.4	Ontology over papers and records; okgg generator decision (§ 9.4)	facets reported; proposals routed to admission only
WP3.11	SPARQL endpoint	the three grounding queries run through it
WP3.12	Consolidation: purpose pack, bloodhound catalogue, kwisatz papers read from the library	no second copy of a P450 value remains

10.2 Phase 3 — Downstream (2/2): domain processes

WP	Work	Done when
WP3.5	Metabolite ID: rules, element set with F/Br/I/Fe, isotope model; N4	precision on a held-out set reported beside chance
WP3.6	LC corroboration	parent/metabolite visibility reported
WP3.7	PGx: PharmVar/CPIC fetch, caller, CYP2D6 CNV/hybrids; N5	CPIC tables reproduced; ClinVar drug response kept
WP3.8	CYP abundance (after U-lev-2)	surrogate peptides mapped by sequence
WP3.9	Spectroscopy readers (EPR, rR, Mössbauer new); N7	minimal probe set of `exp_04_panel` reproduced on records
WP3.10	DDI and clearance; N6	FDA index pairs recovered

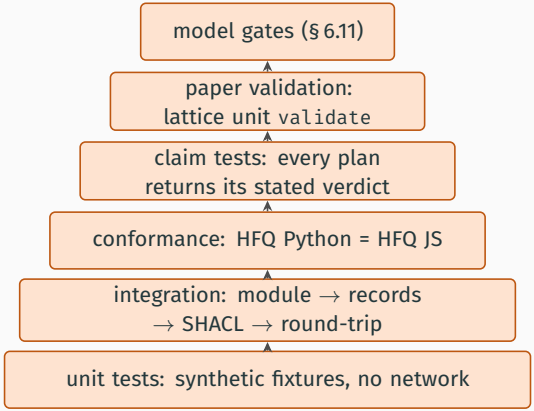
Phase 3 has no internal order: every WP waits only for WP2.4 (and WP3.8 also for U-lev-2).

10.2 Phase 4 — Model

WP	Work	Done when
WP4.1	Corpus builder; licence filter	every line traceable to a plan, run, record or paper section
WP4.2	Baseline Qwen2.5 0.5B LoRA (lattice unit `train`)	trained adapters on `lattice-results/train`
WP4.3	Evaluation harness (§ 6.11)	metrics reported per seed
WP4.4	TIES merge; decide size	the gates in § 6.11 pass
WP4.5	Serve on AppHubAI; dsl-writer integration; N8; the introduction	dsl-writer uses the model for `hfq`; repair rate tracked

Phases gate on acceptance criteria, not on calendar.

11 Verification, from fixtures to model gates



- Tests never touch the network. Each module is tested on synthetic fixtures, following proteomics' `tests/synthetic.py`.
- Every number on the front page is checked against the records it reads from (R12).
- `render_reference.py --check` fails if the specification's reference has drifted from `components.json`; `web`'s `npm run check` fails if any edge does not resolve.
- Negative controls are kept beside every positive result, as in proteomics (chance precision) and admission (random controls).

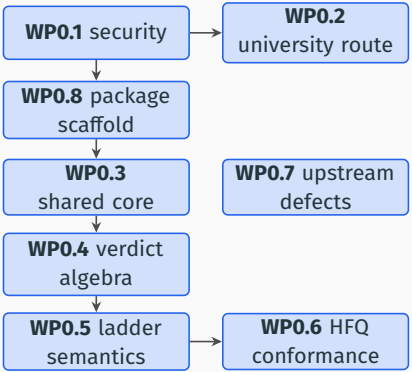
12.1 Risks

Risk	Consequence	Mitigation
Credential in public history	account misuse	WP0.1 before anything else
Licences of KEGG, PharmVar, instrument data	redistribution violation; tainted model	§ 9.3 projections; licence filter in WP4.1
AppHub may not reach external APIs	`fetch` cannot run there	run `fetch` on the laptop, push snapshots through Gitea
Three HFQ copies drift	different answers by runtime	WP0.6 conformance in CI
Framework quantities read as clinical values	misleading report	R6, R12, basis on every CKG fact (§ 6.10)
Substrate-scope questions	the model invents scope	R9; refusal examples in the corpus
raw2dcat source missing on `main`	that code cannot be reused	ask the user (D9)
Scope creep across 102 components	nothing finishes	phases gate on acceptance criteria, not on calendar

12.2 Decisions

#	Decision	Default if not decided
D1	`dcat-ap-plus-p450` location	**decided:** `levinthal/dcat-ap-plus-p450`
D2	Where the shared core lives	inside this package until a second extension adopts it
D3	Stop citing `cytochrome/sources/` as the foundation	yes; ground on `models/sources/` and the ladder papers
D4	The library's position on substrate scope	record, never infer (R9)
D5	First-release sources	the open-licence set in § 6.2
D6	Model size and AppHub budget	0.5B baseline, then decide
D7	Rewrite `cytochrome-module` in place or add a `p450` module	add a `p450` module; keep the current one as the transaminase demo
D8	Verdict for a rungless ladder	`empty`, as the paper says
D9	Where raw2dcat's `kg/federation` source is	—

What happens next: Phase 0



1. **WP0.1** first: untrack the file, rotate the credential.
2. **WP0.2** university route; **WP0.8** scaffold.
3. **WP0.3** shared core, then **WP0.4** verdict algebra.
4. **WP0.5** `subfloor`, rungless ladder; unblocks N2.
5. **WP0.6** HFQ conformance; unblocks WP2.5.
6. **WP0.7** upstream defects; U-lev-2 gates WP3.8.

Open for review: decisions D2–D9 (§ 12.2), and what the specification marks as proposals: the verdict algebra (§ 7.1), the class sketch and the witness sets (§ 6.5).